Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3481-WE01

Title: MEng Engineering, BEng Engineering

Power Semiconductor Devices

Time Allowed:	2 hours	
Additional Material provided:	Generic physical constants:- $q = e = 1.60 \times 10^{-19}$ C 1 eV = 1.60×10^{-19} J $m_e = 9.1 \times 10^{-31}$ kg $h = 6.63 \times 10^{-34}$ J s $c = 3.00 \times 10^{8}$ m s^{-1} $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} = 8.85×10^{-14} F cm^{-1} $k = k_B = 1.38 \times 10^{-23}$ J K^{-1} = 8.6×10^{-5} eV K^{-1} $N_A = 6.02 \times 10^{23}$ $mole^{-1}$	
Materials Permitted:	Open books / open notes.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	All relevant working must be shown. Only authorised calculators should be used.

Revision:	

Question 1

A thin film of gallium antimonide (GaSb) is doped with acceptors only such that the Fermi level is exactly half way between the intrinsic Fermi level and the valence band maximum, at $T = 300\ K$. *Data for GaSb is provided at the end of the question.*

(a) (i) Determine the type (*n*, *p* or *intrinsic*) of the semiconductor.

(ii) Calculate the density of majority and minority carriers.

[20%]

The film is now illuminated from the top, as shown in **Figure Q1**, with a focused beam that generates a steady-state excess of electrons at $x = 0$ of $\delta n = 10^{12}\ cm^{-3}$. The excess carriers are uniformly distributed in the plane normal to the x axis. The film thickness is $1\ \mu m$ and its cross-section area is $0.01\ cm^2$. Use these steady state conditions for sections (b) and (c).

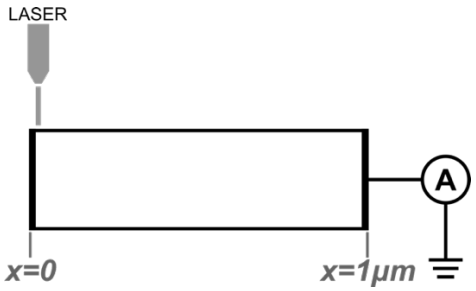


Figure Q1: Cross section of a thin GaSb film and its measurement set-up

(b) (i) Show that the film is much thinner than the electron diffusion length L_n.

(ii) Given your findings in section (b)(i), the recombination term of the continuity equation can be neglected. Derive an expression for the excess of carriers in the x direction of the film $\Delta n(x)$, if the electrode at $x = 1\ \mu m$ is measuring a constant current of $-0.6\ mA$.

[40%]

(c) (i) Explain the difference between steady-state conditions and equilibrium.

Is the semiconductor film in equilibrium, under the conditions in section (b)?

(ii) Calculate the values of the quasi-Fermi level for electrons with respect to the intrinsic Fermi level, $F_n - E_i$, at $x = 0$ and $x = 1\ \mu m$.

(iii) Draw a schematic sketch of the band diagram of the semiconductor, noting the valence and conduction band edges, the electron and hole quasi-Fermi levels and the intrinsic Fermi level. Focus on a qualitative depiction rather than numerical accuracy.

continued

If you did not solve section (b) you may use:

$p_0 = 10^{16}\ cm^{-3}$, $n_0 = 5 \times 10^{8}\ cm^{-3}$, $\Delta n(0) = 10^{13}\ cm^{-3}$, $\Delta n(1\ \mu m) = 10^{11}\ cm^{-3}$.

[40%]

Data for GaSb:

$$E_g = 0.72\ eV \quad m_n^* = 0.04\ m_e \quad m_p^* = 0.8\ m_e \quad \mu_n = 3{,}000\ \frac{cm^2}{V \cdot s}$$

$$\mu_p = 1{,}000\ \frac{cm^2}{V \cdot s} \qquad \tau_n = 2 \times 10^{-7}\ s \qquad \tau_p = 5 \times 10^{-8}\ s$$

Question 2

(a) (i) Explain what a direct band gap is. Will a material with a direct band gap be efficient both as a light emitting device and a light sensitive device?

(ii) Gallium arsenide (GaAs) has a direct band gap. Will a photodetector built from GaAs be sensitive for (1) fibre optics communications wavelength: $\lambda_{IR} = 1.55\ \mu m$; (2) visible green wavelength: $\lambda_{green} = 550\ nm$; (3) short UV (UV-C) wavelength: $\lambda_{UV} = 200\ nm$?

Data for GaAs is provided at the end of the question.

[12%]

The band structure diagram in **Figure Q2** is that of a GaAs-based pn junction diode in equilibrium, with a surface area of $1\ cm^2$. Its built-in potential is $V_{bi} = 1.23\ V$.

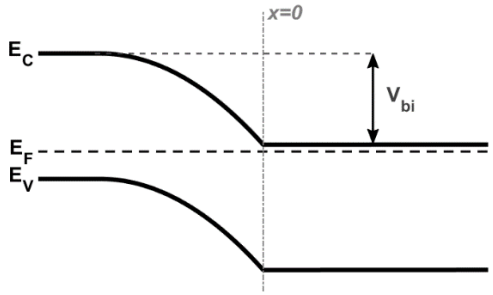


Figure Q2: Band structure of a GaAs pn junction diode

(b) In reverse bias of $V_r = 1.3\ V$, the measured junction capacitance is $C_r = 3.3\ nF$. Calculate the doping in both sides of the junction.

[28%]

(c) The light doping in one side of the junction means that the series resistance of the quasi-neutral region cannot be ignored. If you did not solve section (2), you may use $N_A = 5 \times 10^{14}\ cm^{-3}$.

(i) Calculate the resistance of the lightly doped quasi-neutral region if its length is $2\ cm$.

(ii) Calculate the saturation current of the diode.

(iii) The diode is connected to a current source supplying a bias of $10mA$. What is the potential drop over the entire device?

[48%]

(d) How would you change the diode's doping profile to transform it into a Zener diode? Draw the modified band diagram and explain which parameters have changed to allow tunnelling current.

[12%]

continued

Data for GaAs:

$$E_g = 1.42\ eV \quad \epsilon_r = 12.9 \quad n_i = 2 \times 10^6\ cm^{-3} \quad \mu_n = 8{,}500\ \frac{cm^2}{V \cdot s}$$

$$\mu_p = 400\ \frac{cm^2}{V \cdot s} \qquad \tau_n = 10^{-9}\ s \qquad \tau_p = 5 \times 10^{-11}\ s$$

continued

Question 3

(a) Calculate the theoretical maximum operating temperature for silicon carbide with $N_D=5\times10^{14}$ cm^{-3}.

[16%]

(b) Calculate the electron mobility at the temperature calculated in part (a).

[16%]

(c) Determine the expected increase in the on-state losses for a diode manufactured with an intrinsic region with $N_D=5\times10^{14}$ cm^{-3} that is 5 μm when the temperature is equal to that determined in part (a).

[24%]

(d) Express how the increase in power losses is influenced by the current density through the device and describe the magnitude of this change in comparison to the change in a Schottky diode with an equivalent current and voltage ratings.

[44%]

continued

Question 4

(a) A power electronic circuit contains a silicon carbide MOSFET which is rated at 650 V. Because of a lack of suitable parts, this MOSFET has been replaced with a 1200 V MOSFET. Determine the change in on-state losses.

[16%]

(b) The Insulated Gate Bipolar Transistor (IGBT) is commonly used in silicon based power electronic circuits. However, it is being replaced by wide bandgap MOSFETs in high performance applications. Explain why this move is enabling power electronic circuits to become physically smaller.

[24%]

(c) Junction Field Effect Transistors (JFETs) have significant potential in high temperature power electronics due to their exceptional ruggedness. Calculate the channel width for a JFET so that it has a threshold voltage that is identical to that of a typical power MOSFET (V_T=1.5 V). Assuming that the channel is 2 μm long, with a dopant concentration of 1×10^{16} cm^{-3}. Assuming that the gate width of the JFET is 1 mm, and the cell pitch is 2 μm, calculate the channel resistance, drift region resistance and hence the total specific on-state resistance of a JFET designed to operate at 100 V when the gate – source voltage is 2.5 V.

[60%]

END